

APPDEV1 Project Report - Trekking Management Application

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AI/LLM usage

The usage of AI roughly estimates to around 10 - 12% for debugging and stylizing the front-end using CSS Bootstrap.

Description

The project is trekking management application. We need to have three separate roles and have a secure login authentication along with specialised features for each role. I have approached this project by building a secure login credentials and created the required features one by one.

I have used AI/LLM for debugging certain errors and to have a proper frontend using CSS Bootstrap. I used AI to get these Bootstrap classes to stylize my website. The usage of AI is around 10-12%

Technologies Used

Backend Framework

Python – Core programming language used to build the application logic.

Flask – Lightweight WSGI web framework used to handle routing, request/response cycles, and session management for the entire application (admin, staff, and user modules).

Flask Extensions

Flask-SQLAlchemy – Used as the Object-Relational Mapper (ORM) to interact with the database using

Python classes/objects instead of writing raw SQL queries. It simplifies defining models (treks, staffs, users, bookings) and performing queries like `.filter_by()`, `.query.all()`, and `.session.get()`.

Database

SQLite – Lightweight, file-based relational database (Trekking.db) used to store trek, staff, user, and booking data. Chosen for its simplicity and ease of setup during development, requiring no separate database server.

Templating Engine

Jinja2 – Flask's built-in templating engine used to dynamically render HTML pages (admin.html, user.html)

with data passed from the backend, including template inheritance (`{% extends "base.html" %}`), conditionals, and loops.

Frontend

HTML5 – Used to structure the content of all web pages.

Bootstrap – CSS framework used for responsive layout, pre-styled components (cards, buttons, tables,

forms), and consistent UI design across the admin and user dashboards.

CSS (Custom Inline Styling) – Used alongside Bootstrap for specific theming (e.g., the #1a3c2e brand color

used across headings and buttons).

Session Management

Flask Sessions – Used to maintain login state for admins, staff, and users (e.g., session["user"], session["admin"]) and to restrict access to protected routes.

Standard Library

datetime – Used to handle and format date/time values such as booking dates and trek schedules.

DB Schema Design



Tables: admin, staff, user, trek, booking

admin – id, name, username, password

staff – id, name, email, phone_no, username, password, status

user – id, name, email, phone_no, username, password, status

trek – id, trek_name, location, difficulty, status, start_date, end_date, duration, total_slots,
available_slots, assigned_staff_id (FK)
booking – id, trek_id (FK), user_id (FK), booking_date, status

Relationships

Staff → Trek (1-to-many)
Trek → Booking (1-to-many)
User → Booking (1-to-many)

The reasoning for such a schema:

Separate tables for admin/staff/user avoid unnecessary/null fields per role.
booking acts as a junction table to support many-to-many between users and treks, while storing
booking
specific data.
status fields enable soft-state changes (blacklist, cancel) without deleting records.
Nullable assigned_staff_id lets treks be created before staff assignment.

API Design

No API has been used in this project.

Architecture

Backend: Flask
Model: SQLAlchemy models — admin, staff, user, trek, booking
View: Jinja2 templates (base.html, admin.html, user.html, staff.html)
Controller: Flask routes in app.py handling requests, session auth, and DB operations
Flow: Browser → Flask Route → SQLAlchemy ORM → SQLite DB → Jinja2 Template →
Response
Access Control: Session-based, role-specific (Admin / Staff / User), each with isolated
dashboards and
permissions.

Features

Admin

Dashboard with trek/booking/user/staff stats
Approve/reject staff signups
Search treks, staff, users by ID
Manage treks & reassign staff
Blacklist staff/users

Staff

View assigned treks
Open/close treks
Update slot availability
Remove participants
User
Signup/login
Browse & filter treks (difficulty, location)
Search trek by ID
Book/cancel trek slots

Video Link:

<https://drive.google.com/file/d/1hpKotMA8nuMPQXgn2ABF44SaPwgUmdt4/view?usp=sharing>